

Step 1:-
for
 $LY = B$

$$AX = B$$

let $UX = Y$
 $LY = B$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 14 \\ 13 \\ 17 \end{bmatrix}$$

$$y_1 = 14$$

Step 2:- $UX = Y$

$$\begin{bmatrix} 1 & 5 & 2 \\ 0 & -9 & 1 \\ 0 & 0 & -5/3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 14 \\ -15 \\ -5/3 \end{bmatrix}$$

$$x_2 = -15 \leftarrow y_1 + y_2 = 13 \Rightarrow 13 - 28 = -15$$

$$3(14) + 14(-15) + y_3 = 17 \Rightarrow y_3 = (-5/3)$$

$$14 - 28x_1 + 5x_2 + 2x_3 = 14$$

$$-9y + 2 = 15$$

$$-5/3 = -5/3$$

check it out
 $\begin{matrix} x_1 = 1 \\ y = 2 \\ z = 3 \end{matrix}$

10th April 2021
Wednesday

Cholesky method

Let $AX = B$

let $A = LL^T$ (ii)

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

find $l_{11}, l_{21}, l_{22}, l_{31}, l_{32}, l_{33}$

By (i) & (ii) $LL^T = B$ (iii)

put $LY = B$ solve and find Y

put $L^T X = Y$

$$L^T X = Y$$

find X

Que. Solve the system by Cholesky method.

$$x + 2y + 3z = 5$$

$$2x + 3y + 2z = 6$$

$$3x + 2y + 8z = -10$$

Soln:- let the given system is

$$AX = B$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 3 & 2 & 8 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 5 \\ 6 \\ -10 \end{bmatrix}$$

let $A = LL^T$ i.e. (ii)

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 3 & 2 & 8 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix} = \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{31}l_{11} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

$$u_2 = a_2 - (a_2 \cdot e_1) e_1 = a_2$$

$$\Rightarrow u_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, e_2 = \frac{u_2}{\|u_2\|}$$

$$\|u_2\| = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3}$$

$$e_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

Also

$$u_3 = a_3 - (a_3 \cdot e_1) e_1 - (a_3 \cdot e_2) e_2$$

$$\Rightarrow u_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$u_3 = a_3 - (a_3 \cdot e_1) e_1 - (a_3 \cdot e_2) e_2$$

$$= a_3 - \left(\frac{1}{\sqrt{2}}\right) e_1 - \left(-\frac{1}{\sqrt{3}}\right) e_2$$

$$\Rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} + \frac{1}{\sqrt{3}} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \end{bmatrix} + \begin{bmatrix} 1/3 \\ -1/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} -1/6 \\ 1/6 \\ 1/3 \end{bmatrix}$$

$$0 - \frac{1}{2} + \frac{1}{3} = \frac{1}{3} - \frac{1}{2} = -\frac{1}{6}$$

$$1 - \frac{1}{2} - \frac{1}{3} = \frac{6-3-2}{6} = \frac{1}{6}$$

$$e_3 = \frac{u_3}{\|u_3\|} = \frac{u_3}{1/\sqrt{6}} = \sqrt{6} u_3$$

$$\text{norm } u_3 = \sqrt{\left(\frac{1}{6}\right)^2 + \left(\frac{1}{6}\right)^2 + \left(\frac{1}{3}\right)^2} = \frac{1}{\sqrt{6}}$$

$$e_3 = \begin{bmatrix} -1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}$$

$$= \sqrt{6} \begin{bmatrix} -1/6 \\ 1/6 \\ 2/6 \end{bmatrix}$$

$$\therefore Q = [e_1 \ e_2 \ e_3] = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{2} & -1/\sqrt{2} & 1/\sqrt{6} \\ 0 & 1/\sqrt{2} & 2/\sqrt{6} \end{bmatrix}_{3 \times 3}$$

length of $e_i = 1$

Orthogonal set

dot product of columns

$$e_1 \cdot e_2 = 0, e_1 \cdot e_3 = 0, e_2 \cdot e_3 = 0$$

$$(e_1)_1 \times (e_2)_1 + (e_1)_2 \times (e_2)_2 + (e_1)_3 \times (e_2)_3 = 0$$

Since e_1, e_2, e_3 are orthonormal

Orthogonal Vectors

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$a_2 \cdot e_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = 0$$

$$1/\sqrt{2} - 1/\sqrt{2} = 0$$

$$1 \neq 3 \text{ so}$$

orthogonal

$$e_1 \cdot e_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = 0$$

$$e_3 \cdot e_2 = \begin{bmatrix} -1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = -1/3$$

$$a_3 \cdot e_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} -1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix} = 1/\sqrt{6}$$

Q is orthogonal
 $\therefore Q^{-1} = Q^T = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{6} & 2/\sqrt{6} \end{bmatrix}$

Step 2:- To find R (upper triangular matrix)

$$R = Q^T A$$

$$= \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{6} & 2/\sqrt{6} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} \sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & \sqrt{3} & -1/\sqrt{3} \\ 0 & 0 & 1/\sqrt{6} \end{bmatrix}$$

So, $A = QR = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \\ 0 & 1/\sqrt{6} & 2/\sqrt{6} \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & \sqrt{3} & -1/\sqrt{3} \\ 0 & 0 & 1/\sqrt{6} \end{bmatrix}$

Answer

to prove

Kaplan's $A = QR = [v_1 | v_2 | \dots]$

$A = QR$, where $Q \in \mathbb{R}^{m \times n}$ is an orthogonal set of vectors $= m \times n$

Let $S = \{x_1, x_2, \dots, x_n\}$ $R =$ upper triangular matrix

Proof: S is called orthogonal if $\langle x_i, x_j \rangle = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$

(i) Q is orthogonal

(ii) Norm of each vector in $S = 1$

How to find Q and R $\rightarrow$ Gram-Schmidt orthogonalization
 $\rightarrow$ Householder
 $\rightarrow$ Givens rotation

Sub 2: Decomposition

(i) LU

(ii) PLU

(iii) Cholesky $\rightarrow$ positive definite matrix

(iv) QR $\rightarrow$ invertible matrix $\rightarrow$ orthogonal matrix $= Q$, $R =$ upper triangular

(v) SVD (Singular value decomposition)

Symmetric ($A^T = A$)

Positive definite matrix

non-singular

$$(A:B) = \begin{bmatrix} 1 & 4 & 9 & 16 \\ 0 & -7 & -17 & -22 \\ 0 & -10 & -34 & -30 \end{bmatrix}$$

$$\begin{aligned} R_2 &\rightarrow R_2 - 4R_1 \\ R_3 &\rightarrow R_3 - 9R_1 \end{aligned} \quad \begin{bmatrix} 1 & 4 & 9 & 16 \\ 0 & -7 & -17 & -22 \\ 0 & -10 & -34 & -30 \end{bmatrix} \quad \begin{aligned} R_3 &\rightarrow 7R_3 - 10R_2 \\ R_2 &\rightarrow 7R_2 + 10R_3 \end{aligned} \quad \begin{bmatrix} 1 & 4 & 9 & 16 \\ 0 & -7 & -17 & -22 \\ 0 & 0 & 2 & 10 \end{bmatrix}$$

$$\begin{aligned} x &= 7 \\ y &= -9 \\ z &= 5 \end{aligned}$$

$$x = 7$$

$$x + 4y + 9z = 16$$

$$-7y - 17z = -22$$

$$-10y - 34z = -30$$

$$y = -9$$

$$z = 5$$

method 2 - Gauss Jordan

Somehow

$$\begin{bmatrix} 1 & 4 & 9 & 16 \\ 0 & -7 & -17 & -22 \\ 0 & -10 & -34 & -30 \end{bmatrix}$$

Diff. from elimination
1. Steps -
2. Same

1. Steps -
2. Same

1. Steps -
2. Same

$$\begin{bmatrix} 1 & 4 & 9 & 16 \\ 0 & -7 & -17 & -22 \\ 0 & -10 & -34 & -30 \end{bmatrix}$$

Sub. diag. mat. of co. var.

$$\begin{bmatrix} 1 & 4 & 9 & 16 \\ 0 & -7 & -17 & -22 \\ 0 & 0 & 2 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 7 & 0 & -5 & 24 \\ 0 & -7 & -17 & -22 \\ 0 & 0 & 2 & 10 \end{bmatrix}$$

$$R_4 \rightarrow 2R_4 + 5R_3$$

$$R_2 \rightarrow 2R_2 + 7R_3$$

$$\begin{bmatrix} 14 & 0 & 0 & 98 \\ 0 & -14 & 0 & 126 \\ 0 & 0 & 2 & 10 \end{bmatrix}$$

$$14x = 98 \rightarrow x = 7$$

$$-14y = 126 \rightarrow y = -9$$

$$2z = 10$$

$$z = 5$$

8th April '24
Monday

Today's

LU decomposition

1. LU decomposition
2. LU decomposition

1. LU decomposition
2. LU decomposition

from

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = LU$$

$$AX = B$$

$$UX = B$$

$$UX = B$$

$$UX = B$$

$$UX = B$$

$$UX = B$$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}, U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix} \quad A = LU$$

Steps: $LUx = B$ Step 1: find U first

① first row of U i.e. u_{11}, u_{12}, u_{13}

② first column of L i.e. l_{11}, l_{21}, l_{31}

③ second row of U i.e. u_{22}, u_{23}

④ $\dots$ L l_{21}, l_{31}

⑤ third row of U develop out?

≠ solve the system eqn by LU

$$x + 5y + z = 14$$

$$2x + y + 3z = 13$$

$$3x + y + 4z = 17$$

Solving-

$$A = \begin{bmatrix} 1 & 5 & 1 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{bmatrix} = LU$$

$$\begin{bmatrix} 1 & 5 & 1 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 14 \\ 13 \\ 17 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & 1 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$\begin{bmatrix} 1 & 5 & 1 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & (l_{21}u_{12} + u_{22}) & (l_{21}u_{13} + u_{23}) \\ l_{31}u_{11} & (l_{31}u_{12} + l_{32}u_{22}) & (l_{31}u_{13} + l_{32}u_{23} + u_{33}) \end{bmatrix}$$

on comparing

$$u_{11} = 1$$

$$u_{12} = 5$$

$$u_{13} = 1$$

$$l_{21}u_{11} = 2$$

$$l_{21} = 2$$

$$l_{21}u_{12} + u_{22} = 1$$

$$2 \times 5 + u_{22} = 1$$

$$u_{22} = -9$$

$$l_{21}u_{12} + u_{22} = 1$$

$$2 \times 5 + (-9) = 1$$

$$l_{31}u_{11} = 3$$

$$l_{31} = 3$$

$$l_{31}u_{12} + l_{32}u_{22} = 1$$

$$3 \times 5 + l_{32}(-9) = 1$$

$$15 + l_{32}(-9) = 1$$

$$l_{32} = -14/9$$

$$l_{31}u_{13} + l_{32}u_{23} + u_{33} = 4$$

$$3 \times 1 + (-14/9)u_{23} + u_{33} = 4$$

$$A = LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & -14/9 & 1 \end{bmatrix} \begin{bmatrix} 1 & 5 & 1 \\ 0 & -9 & -1 \\ 0 & 0 & -5/9 \end{bmatrix}$$

By Gram-Schmidt. process, an orthonormal basis

$$W^T = \begin{bmatrix} 1/56 & 3/56 & 1/66 \\ 1/56 & 7/65 & 0 \\ 3/56 & 2/30 & -5/30 \end{bmatrix}$$

$$E = \begin{bmatrix} \sqrt{12} & 0 & 0 \\ 0 & 5/10 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A_{2 \times 3} = E_{2 \times 3}$$

$$A = U \cdot E \cdot V^T$$

where E = diag matrix with singular values (Singular test)

Part			
A	B	C	Complex
1	0	0	Complex
0	5/3	0	Complex

Re. complete system Matrix factor

Singular matrix is in $\mathbb{R}^{m \times n}$
 If inverse do these with it
 is diagonal in nature

Meaning eigen $\lambda_1 = 10, \lambda_2 = 12$

Let $\lambda_1 = 0, \lambda_2 = 10, \lambda_3 = 12$

$$U, V = (\lambda_1 = 17, \lambda_2 = 19) \text{ } E$$

Hint: matrix $A = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$

Step 1: $A \cdot A^T$ matrix over eigen value from U and V

Step 2: $A^T \cdot A$ matrix over eigen value from U and V

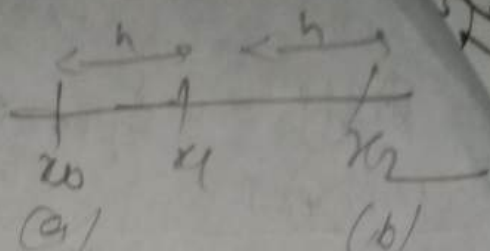
$$U = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \rightarrow V^T = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$

$$E = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

$$A = U \cdot E \cdot V^T$$

Simpson's $1/3$ Rule $\rightarrow$

$$I = \int_a^b f(x) dx$$



$$I = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] \quad \text{--- (1)}$$

Comp for n div in eqn (1)

$$I = \frac{h}{3} [f(x_0) + 4 \{f(x_1) + f(x_3) + \dots + f(x_{2N-1})\} + 2 \{f(x_2) + f(x_4) + \dots + f(x_{2N})\}]$$

Que find $\int_0^1 \frac{1}{1+x} dx$

$\rightarrow$ 2 values

$\rightarrow$ 8 sub-int

$N=$

$N=4$

now Comp for $\int_{-1}^1 f(x) dx$

Gauss Legendre - Integral Rule :-

$$I = \int_{-1}^1 f(x) dx$$

limit $[-1, 1]$

$w(x) = 1$ (weight fun)

two point $\int_{-1}^1 f(x) dx = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$

now limit $[-1, 1]$ to $[a, b]$ transform

$$\int_a^b f(x) dx \xrightarrow{\text{transform}} \int_{-1}^1 f(t) dt$$

$$x = \frac{1}{2} [(b-a)t + (b+a)]$$

$f(x) = f(t)$

$$dx = \frac{(b-a)}{2} dt$$

remark $\frac{1}{\sqrt{3}} = 0.5773$

$S_3 = 1.7320$

3 deg
Three point

$$\int_{-1}^1 f(x) dx = \frac{5}{9} f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9} f(0) + \frac{5}{9} f\left(\sqrt{\frac{3}{5}}\right)$$

$$x_0 = \frac{5}{9} \frac{\sqrt{3}}{5} = 0.55, \quad \sqrt{\frac{3}{5}} = 0.7741, \quad \frac{8}{9} = 0.888$$

Gauss Hermit

$$\int_{-\infty}^{\infty} f(x) dx = \frac{\sqrt{\pi}}{2} \int_{-\infty}^{\infty} e^{-x^2} f(x) dx$$

① two point $\int_{-\infty}^{\infty} e^{-x^2} f(x) dx = \frac{\sqrt{\pi}}{2} \left[f\left(-\frac{1}{\sqrt{2}}\right) + f\left(\frac{1}{\sqrt{2}}\right) \right]$

② Gauss three point $\int_{-\infty}^{\infty} e^{-x^2} f(x) dx = \frac{\sqrt{\pi}}{6} \left[f\left(-\frac{\sqrt{6}}{2}\right) + 4f(0) + f\left(\frac{\sqrt{6}}{2}\right) \right]$

Remark: $\frac{\sqrt{\pi}}{2} = 0.8862, \quad \frac{\sqrt{\pi}}{6} = 0.2954, \quad \frac{1}{\sqrt{2}} = 0.7071$
 $\frac{\sqrt{6}}{2} = 1.2247$

for Differentiation using Newton's formula Δ

$$f''(x) = \frac{d^2 y}{dx^2} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 - \frac{5}{6} \Delta^5 y_0 + \dots \right]$$

$$f'(x) = \frac{dy}{dx} = \frac{1}{h} \left[\Delta y_0 + \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 + \frac{1}{4} \Delta^4 y_0 + \dots \right]$$

Notes:-

by backward diff. oper $\frac{2}{h}$ term all in $2h(x)$

$$f''(x) = \frac{d^2 y}{dx^2} = \frac{1}{h^2} \left[\Delta^2 y_0 + \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 + \frac{5}{6} \Delta^5 y_0 + \dots \right]$$

$$f'(x) = \frac{dy}{dx} = \frac{1}{h} \left[\Delta y_0 + \frac{1}{2} \Delta^2 f(x_0) + \frac{1}{3} \Delta^3 f(x_0) - \frac{1}{4} \Delta^4 f(x_0) + \dots \right]$$

22nd April 2019
Monday

Singular Value Decomposition (SVD)

$$A_{m \times n} = U_{m \times m} \Sigma_{m \times n} V^T_{n \times n}$$

$U_{m \times m}$ & $V_{n \times n}$ are unitary matrices which are orthogonal in $\mathbb{R}$ space.
 $\Sigma_{m \times n}$: Single entry value
 $V^T_{m \times n} \rightarrow$ orthogonal
 $U_{m \times n} \rightarrow$ orthogonal
 $U \cdot U^T = I$
 $V \cdot V^T = I$ (Identity matrix)
 Single value

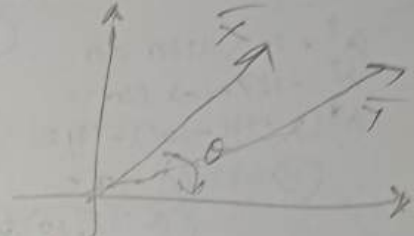
$$A^T \cdot A = (V \Sigma^T U^T) U \Sigma V^T$$

$$A^T A = V \Sigma^T \Sigma V^T$$

$$A \cdot A^T = (U \Sigma V^T) \cdot (V \Sigma^T U^T)$$

$$A \cdot A^T = U \Sigma \Sigma^T U^T$$

Rotation and stretching
 Gram-Schmidt



rotation, stretch, rotation
 $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \rightarrow$ Unitary transformation
 $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

Que $A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}_{2 \times 3}$

4th eval on vectors

$$A^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix}_{3 \times 2}$$

8x92

$$A \cdot A^T = \begin{bmatrix} 11 & 1 \\ 1 & 11 \end{bmatrix}_{2 \times 2}$$

$$A^T \cdot A = V$$

$$A \cdot A^T = U$$



Stretching



ellipse

$$(A - \lambda I) = 0$$

$$\begin{bmatrix} (11-\lambda) & 1 \\ 1 & (11-\lambda) \end{bmatrix} = 0$$

$$(11-\lambda)^2 - 1^2 = 0 \rightarrow (A+I)(A-I) = 0$$

$$(11-\lambda) = 22\lambda + 20$$

$$(11-\lambda)(11-\lambda-1) = 0$$

$$(12-\lambda)(10-\lambda) = 0$$

eigen values $\lambda_1 = 10$
 $\lambda_2 = 12$

for $\lambda_1 = 10$ 2 eigen vectors

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\sqrt{2} + 1 = \sqrt{2} \Rightarrow \sqrt{2} - \sqrt{2} = 0$$

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Gram-Schmidt

orthogonal

$$A = U \cdot \Sigma \cdot U^T$$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} = U$$

① Difference Operator :-

Central

FORWARD

BACKWARD

$\delta f(x_i)$

$$\Delta f(x_i) = f(x_{i+1}) - f(x_i)$$

$$\nabla f(x_i)$$

$$\Delta^2 f(x_i)$$

$$\Delta^3 f(x_i)$$

$h = (x_i - x_{i-1})$ is constant, $\Delta f(x_i)$ is constant

② Numerical Integration :-

equi-spaced

Discrete Difference

unequ-spaced

Newton's

Cotes

Cotes

Uncon. Dist

Gauss Quadrature

Trapezium Rule

Simpson's 1/3 Rule

Gauss Legendre Rule

Gauss Hermite Rule

Trapezium Rule :- $\rightarrow$

$h = \frac{b-a}{N}$ for getting accuracy every point Trapezoidal rule :-

$$I = \int_a^b f(x) dx$$

$$I \approx \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{N-1}) + f(x_N)]$$

Que find $I = \int_0^1 \frac{1}{1+x} dx$ & $N=8$

$$h = \frac{1-0}{8} = \frac{1}{8}$$

$$\text{Exact sol} = I = \ln 2 = 0.693147$$

$$\text{error} = \text{Exact sol} - I_{\text{approx}}$$

Stage 1: $A = \begin{bmatrix} 3 & 1 & 1 \\ -1 & 3 & 1 \end{bmatrix}$, $A^T = \begin{bmatrix} 3 & -1 \\ 1 & 3 \end{bmatrix}$, $A \cdot A^T = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$

$A^T \cdot A = \begin{bmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 10 \end{bmatrix}$ 3x3

$(A - \lambda I) = 0$

$\lambda_1 = \text{minor of } a_{11}$

$\rightarrow \begin{bmatrix} (10-\lambda) & 0 & 2 \\ 0 & (10-\lambda) & 4 \\ 2 & 4 & (10-\lambda) \end{bmatrix} \rightarrow \begin{bmatrix} \lambda^3 - 3\lambda^2 + 2\lambda - 5 \end{bmatrix}$

find char eqn

$\lambda^3 - 3\lambda^2 + 2\lambda - 5 = 0$

$\lambda(\lambda^2 - 3\lambda + 2) = 0$

$\lambda^2 - 3\lambda + 2 = 0$

$\lambda^2 - 2\lambda - 10\lambda + 20 = 0$

$\lambda^2(\lambda - 2) - 10(\lambda - 2) = 0$

$(\lambda - 2)(\lambda - 10) = 0$

$\lambda = 2, 10$

$\lambda_1 = 2$

$\lambda_2 = \text{minor}$

$\begin{vmatrix} 10 & 4 \\ 4 & 2 \end{vmatrix} = 4$, $\begin{vmatrix} 10 & 2 \\ 2 & 2 \end{vmatrix} = 0$

$4 + 16 + 100 = 120$, $\begin{vmatrix} 10 & 0 \\ 0 & 10 \end{vmatrix} = 100$

$V^T = \begin{bmatrix} v_1^T \\ v_2^T \\ v_3^T \end{bmatrix}$ singular value

By Cramer's Rule,

$\lambda_1 = 12$

$\lambda_2 = 10$

$V = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$\begin{bmatrix} 2 & 0 & 2 \\ 0 & 2 & 4 \\ 2 & 4 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$

$(A - \lambda I)/x = 0$

$\begin{bmatrix} -2 & 4 \\ 4 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$\begin{bmatrix} -2 & 2 \\ 2 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$\begin{bmatrix} 0 & -2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

$V = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & 0 & -5 \end{bmatrix}$ 3x3

$V^T = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & -5 \end{bmatrix}$

$\sqrt{2^2 + 1^2} = \sqrt{5}$
 $\sqrt{4^2 + 1^2 + 16} = \sqrt{21}$
 $\sqrt{1^2 + 4^2} = \sqrt{17}$

$\begin{bmatrix} 0 & 0 & 2 \\ 0 & 0 & 4 \\ 2 & 4 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\lambda_3 = 0$

$\begin{bmatrix} 10 & 0 & 2 \\ 0 & 10 & 4 \\ 2 & 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\frac{x_1}{-0.6} = \frac{-x_2}{1} = \frac{x_3}{0}$

$\frac{x_1}{4} = \frac{-x_2}{-8} = \frac{x_3}{-20}$

$(l_{11})^2 = 1 \Rightarrow l_{11} = 1$ | $(l_{21})^2 + (l_{22})^2 = 8 \Rightarrow \begin{cases} l_{22} = 4 \\ l_{21} = 2 \end{cases}$
 $l_{31}l_{32} = 2 \Rightarrow l_{32} = 2$
 $l_{31}l_{33} = 3 \Rightarrow l_{33} = 3$ | $l_{21}l_{22} + l_{23}l_{24} = 2 \times 4 + 2 \times 3 = 14$
 $\therefore L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix}$ $AX = B \Rightarrow A = LL^T$ (2)

(1) + (2) $LL^T X = B \Rightarrow$ (3)
 put $L^T X = Y$ where $Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$
 from (1) becomes

$LY = B$
 $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 2 & 0 \\ 3 & 8 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \\ -10 \end{bmatrix}$ $\Rightarrow y_1 = 5$
 $2y_1 + 2y_2 = 6 \Rightarrow 2y_2 = 6 - 10 = -4 \Rightarrow y_2 = -2$
 $3y_1 + 8y_2 + 3y_3 = -10 \Rightarrow 3(5) + 8(-2) + 3y_3 = -10 \Rightarrow 15 - 16 + 3y_3 = -10 \Rightarrow 3y_3 = -9 \Rightarrow y_3 = -3$

from (4) $L^T X = Y$
 $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 8 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \\ -3 \end{bmatrix}$

$2z = -3 \Rightarrow z = -3/2$
 $2y - 2 = -2 \Rightarrow 2y = 0 \Rightarrow y = 0$

$x + 2y + 3z = 5 \Rightarrow x + 0 + 3(-3/2) = 5 \Rightarrow x - 9/2 = 5 \Rightarrow x = 5 + 9/2 = 19/2$

13th April 2024
 Saturday

QR Decomposition

any matrix $A = QR$

where $Q =$ orthogonal matrix of $n \times n$
 $R =$ upper triangular matrix of $n \times n$
 Inverse $R^{-1} = n \times n$

Now to find Q ?

Step (i) let column of A be $a_1, a_2, \dots, a_n$
 i.e. $A = [a_1 \ a_2 \ \dots \ a_n]$

Step (ii) Use Gram-Schmidt process to make the column of A into orthonormal vectors of following:
 Define new vectors $u_1, u_2, \dots, u_n$ using column of A

$$u_1 = a_1 \Rightarrow e_1 = \frac{u_1}{\|u_1\|} \quad \text{length of } u_1 \text{ is } \|u_1\|$$

$$u_2 = a_2 - (a_2 \cdot e_1)e_1 \Rightarrow e_2 = \frac{u_2}{\|u_2\|} = \text{unit vector}$$

$$u_3 = a_3 - (a_3 \cdot e_1)e_1 - (a_3 \cdot e_2)e_2 \Rightarrow e_3 = \frac{u_3}{\|u_3\|}$$

$$u_n = a_n - (a_n \cdot e_1)e_1 - (a_n \cdot e_2)e_2 - \dots - (a_n \cdot e_{n-1})e_{n-1}$$

Here $e_1, e_2, \dots, e_n$ are pairwise orthogonal Normal vectors $e_n = \frac{u_n}{\|u_n\|}$

Sign of e_i $[e_1 \ e_2 \ \dots \ e_n] \Rightarrow Q^{-1} = Q^T$

OrthoNormal
 (unit vectors) $\|u_i\| = 1$
 $(u_i \cdot u_j) = \delta_{ij}$ orthonormal

eg: $e_1 = (1, 0, 0)$
 $e_2 = (0, 1, 0)$
 $e_3 = (0, 0, 1)$

How to find R:-

we have $A = QR$

pre multiply on the both side Q^T

$$Q^T(A) = Q^T(QR) = IR = R$$

Thus $R = Q^T A = (Q^T A)$

Now unit vectors are e_1, e_2, e_3

Finally express the matrix A as $A = QR$

eg:- find QR decomp. of matrix $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

Gram-Schmidt process
 let $A = [a_1 \ a_2 \ a_3]$
 Thus $a_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, a_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, a_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

$$A = QR$$

Step:- find Q using Gram-Schmidt process
 Now, define u_1, u_2, u_3

$$u_1 = a_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$\|u_1\| = \|a_1\| = \text{length of } a_1 = \sqrt{(1)^2 + (1)^2 + (0)^2} = \sqrt{2}$

$e_1 = \frac{u_1}{\|u_1\|} = \frac{u_1}{\sqrt{2}} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$